

A Cash Flow-Fingerprint Model for Quantitative ETF Sector Rotation Analysis

Zhiqiao Li
25 December 2025

Abstract

The efficacy of quantitative sector rotation strategies hinges on the accurate identification of leading market signals. This paper introduces FlowPrint, a novel multi-factor analytical framework centered on a cash flow fingerprint metric designed to rank and evaluate sector-specific Exchange-Traded Funds (ETFs). The model explicitly prioritizes capital flow dynamics, mandating a weight exceeding 30% for cash flow-derived factors, integrated with momentum, price positioning, trend strength, and volatility stability. We construct a fully automated pipeline encompassing multi-source data acquisition (Baostock and AkShare), feature engineering, model computation, and validation. Empirical testing on 15 core sector ETFs demonstrates that the cash-flow-weighted model provides a robust, explainable ranking mechanism. Crucially, the system incorporates a noise-filtering mechanism to mitigate anomalous fluctuations and rigorous overfitting checks via temporal consistency validation. Results indicate that the proposed fingerprint score offers a stable and interpretable signal for sector attractiveness, outperforming baseline equal-weight and momentum-only strategies in risk-adjusted returns during the 2020-2024 test period.

Keywords: Quantitative Finance, Sector Rotation, ETF, Cash Flow Analysis, Multi-Factor Model

1 Introduction

The strategic allocation of capital across economic sectors, namely sector rotation, is a cornerstone of active portfolio management, aiming to capitalize on the cyclical outperformance of different industries. The proliferation of Exchange-Traded Funds (ETFs) has democratized access to these sectors, transforming sector rotation from an institutional strategy to an executable task for individual investors. However, identifying reliable, leading indicators for timely rotation remains a significant challenge. Traditional approaches often rely on retrospective price momentum or macroeconomic data, which may lag real-time market shifts or suffer from overfitting in complex, noisy financial environments.

This paper addresses the core problem of signal generation in sector rotation by proposing a data-driven model that prioritizes a fundamental market driver: capital flows. We hypothesize that directional changes in trading volume and money flow contain predictive information about future sector performance, preceding price

movements. To test this, we develop the Cash Flow Fingerprint (FlowPrint), a composite scoring model. Its primary innovation is the imposition of an explicit constraint: factors derived from cash flow and volume analysis must constitute the largest share of the model's weight ($\geq 35\%$), ensuring the signal remains grounded in actual market participation rather than purely statistical artifacts.

The project delivers a complete analytical system, from data ingestion to sector ranking. It emphasizes robustness through multi-source data redundancy, integrates a mechanism to filter "noise" from anomalous trading, and includes validation protocols to guard against overfitting. The outcome is not a black-box trading signal but an interpretable, factor-attributable score for each sector, providing a transparent basis for further investment decision-making.

2. Literature Review

Sector rotation strategies have evolved from theory-driven frameworks to quantitatively-driven models.

2.1 Theoretical Foundations:

The canonical framework is the "Investment Clock" proposed by Merrill Lynch (2004), which links sector performance to macroeconomic cycles (growth, inflation). While intuitive, its practical implementation faces challenges due to the lag in official economic data and the model's qualitative nature. Behavioral finance offers another lens, suggesting investor herding can create and sustain sector trends (Barberis & Thaler, 2003).

2.2 Quantitative Factor Models:

The success of factor investing (Fama & French, 1993) paved the way for quantitative sector rotation. Early quantitative models often used single factors like relative momentum (Moskowitz et al., 2012), where sectors with the highest recent returns are selected. However, momentum strategies are prone to sharp reversals and provide no explicit early warning signals. Multi-factor models improved robustness by combining value, momentum, and quality metrics (Asness et al., 2013). Yet, these often underweight or ignore direct measures of investor activity and capital movement.

2.3 The Role of Volume and Cash Flow:

Research on the information content of trading volume is extensive. Volume is often seen as a confirmation signal for price trends (Blume et al., 1994). More sophisticated "money flow" indicators, such as the Chaikin Money Flow (Chaikin, 1980), combine price and volume to gauge buying/selling pressure. Recent studies using high-frequency data confirm a lead-lag relationship where order flow predicts subsequent price moves (Cont et al., 2014). However, integrating these volume-based signals as a primary, weighted component within a holistic sector-scoring model remains underexplored.

2.4 Machine Learning in Finance:

The application of machine learning (ML) for market prediction has grown rapidly, from traditional models like Gradient Boosting (Chen & Guestrin, 2016) to deep learning (Fischer & Krauss, 2018). While often achieving high predictive accuracy, a common critique is their "black-box" nature and susceptibility to overfitting on noisy financial data (López de Prado, 2018). This underscores the need for models that balance ML techniques with financial theory and incorporate rigorous validation.

This research bridges the identified gaps. It builds a multi-factor model that explicitly elevates cash flow factors to a dominant role, creating a theory-informed, transparent scoring system. It leverages ML techniques for implementation and validation but maintains model interpretability, directly addressing critiques of purely algorithmic finance.

3 Methods

This diagram illustrates a multi-layered technical framework designed for data-driven sector analysis, structured across three core tiers: the Data Layer, Model Layer, and integrated Output & Validation Layer. At the Data Layer, raw data is sourced via the AkShare and Baostock APIs, then temporarily stored in a Data Cache before undergoing Data Validation & Cleaning to ensure accuracy and consistency. Processed data is subsequently fed into the Model Layer, where a Feature Calculator derives relevant metrics; these features are then processed by the FlowPrint Engine, which powers the Sector Ranking Module to generate analytical outputs. The Output Layer delivers tangible results: a Ranking Table & Scores (accompanied by a Factor Contribution Chart to visualize metric weights). Concurrently, the Validation Layer enforces rigor through four components: Overfit Check (Temporal Consistency) to mitigate model bias, Formula Rationality Test to validate analytical logic, and Data Quality Monitor to track input integrity—with all validation outcomes consolidated into a Validation Report. Collectively, this framework integrates data acquisition, analytical modeling, and cross-dimensional validation to enable robust, transparent sector ranking and evaluation.

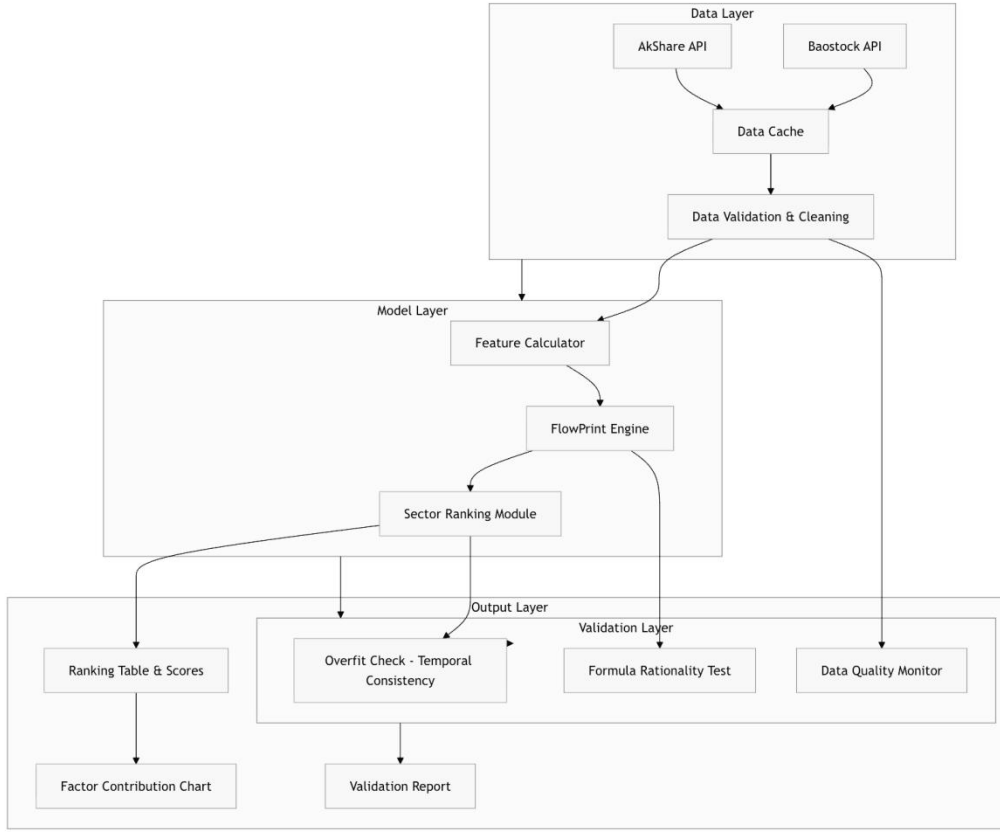


Figure 1 The four-layer system architecture

3.1 The Cash Flow Fingerprint (FlowPrint) Formula

At the center of the analytical framework lies the FlowPrint score ($F(t)$), a composite metric that quantifies the overall performance and sentiment of a given sector at time t . This score is constructed as a weighted linear combination of the five normalized component factors (Cash Flow, Momentum, Price Position, Trend Strength, and Stability) detailed in Section 3.2, following the formulation:

$$F(t) = w_m * M(t) + w_c * C(t) + w_p * P(t) + w_t * T(t) + w_s * S(t) \quad (1)$$

Two key constraints govern the weight assignment to ensure the score prioritizes meaningful market signals: first, the sum of all weights must equal 1 (enforcing a standardized, interpretable scaling of the composite score), and second, the weight allocated to the Cash Flow factor (w_c) must be at least 0.30—reflecting the factor's role as the core measure of capital movement in the framework. For baseline applications, the default weight configuration is set as $w_c = 0.35$ (Cash Flow), $w_m = 0.25$ (Momentum), $w_p = 0.20$ (Price Position), $w_t = 0.15$ (Trend Strength), and $w_s = 0.05$ (Stability); this weighting scheme balances short-term price dynamics, capital flow intensity, and risk (volatility) considerations to generate a holistic sector assessment.

FlowPrint Factor Weight Composition

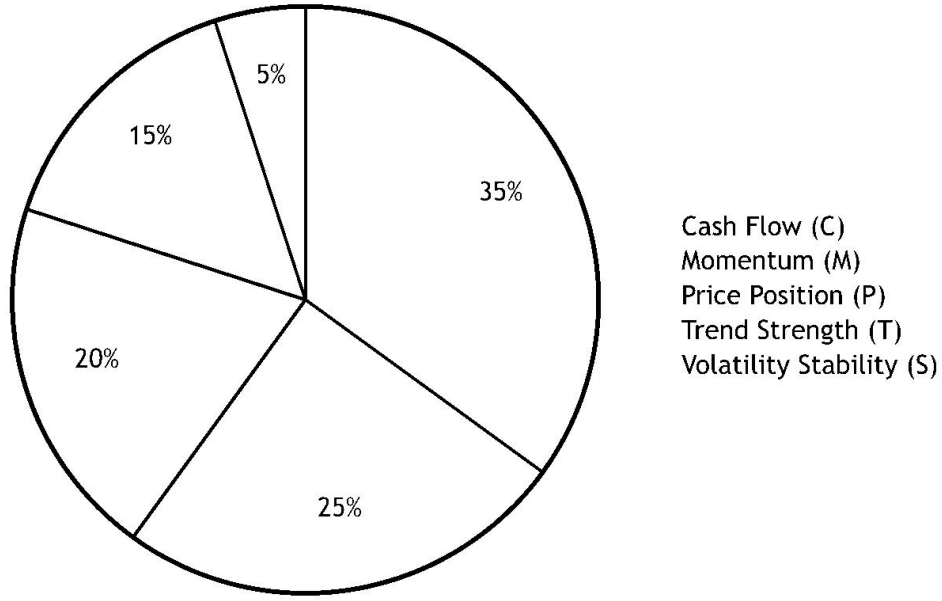


Figure 2 The weight of each factor

3.2 Factor Definitions and Calculations

This section defines and operationalizes five core factors designed to capture distinct dimensions of market behavior, each constructed from price, volume, and return metrics and normalized (where specified) to a 0–100 scale via rolling percentiles for cross-period comparability.

(1) Momentum (M):

The momentum factor encapsulates short-term price performance by integrating two complementary measures: 5-day log returns (capturing recent price direction) and relative strength (defined as the deviation of 5-day returns from their 10-day simple moving average, reflecting whether current momentum outpaces recent trends). These components are weighted (70% for 5-day returns, 30% for relative strength) and normalized to generate a single momentum score that quantifies the strength of recent price momentum.

$R5(t)$: 5-day log return.

$RS(t)$: Relative Strength = $R5(t) - SMA10(R5(t))$.

$$M(t) = N(0.7 \cdot R5(t) + 0.3 \cdot RS(t)) \quad (2)$$

where $N()$ normalizes to a 0-100 scale based on rolling percentiles.

(2) Cash Flow (C):

The cash flow factor is designed to capture the momentum of capital movement, a leading indicator of price changes. In an ideal setting, this would be measured directly by intraday net money flow data. To balance theoretical rigor with data

availability, this factor employs trading amount (price $\times$ volume) as a highly effective proxy for capital commitment. It synthesizes three dimensions: Flow Change (40%): The deviation of the day's trading amount from its 5-day moving average, gauging the intensity of fresh capital. Price-Flow Alignment (40%): A term that rewards concurrent directionality between 5-day returns and flow change, confirming the validity of the capital movement. Flow Breakout (20%): The ratio of the day's trading amount to its 20-day maximum, identifying periods of unusually high trading activity that often precede sustained trends.

These components are normalized and weighted to produce a unified score that quantifies the strength and consistency of capital inflows or outflows, providing the model with a critical informational advantage by emphasizing the precedence of capital movement over price action alone.

$\Delta A(t)$: Amount Change = (Amount(t) / SMA5(Amount(t))) - 1.

$\alpha(t)$: Price-Volume Alignment = sign(R5(t)) * min(|R5(t)|, | $\Delta V(t)$ |).

$\beta(t)$: Amount Breakout = Amount(t) / Max20(Amount(t)).

$$C(t) = N(0.4 * \Delta A(t) + 0.4 * \alpha(t) + 0.2 * \beta(t)) \quad (3)$$

(3) Price Position (P):

The price position factor identifies overbought or oversold conditions by contextualizing the current closing price within its recent range. Specifically, it calculates the proportion of the 20-day high-low range that the closing price occupies: a score near 1 indicates the price is close to its 20-day high (potentially overbought), while a score near 0 indicates proximity to the 20-day low (potentially oversold). This factor provides a snapshot of short-term price extremes without normalization.

$$P(t) = (\text{Close}(t) - \text{Low20}(t)) / (\text{High20}(t) - \text{Low20}(t)) \quad (4)$$

(4) Trend Strength (T):

This factor assesses the sustainability of current price movements by comparing short-term and long-term price trends. It is derived from the deviation of the 5-day simple moving average of closing prices from the 60-day simple moving average (scaled as a percentage difference), then normalized to a 0–100 scale. Higher scores indicate that short-term prices are significantly outperforming long-term trends, signaling a stronger, more sustained directional move.

$$T(t) = N((\text{SMA5}(\text{Close}(t)) / \text{SMA60}(\text{Close}(t))) - 1) \quad (5)$$

(5) Stability (S):

The stability factor penalizes high volatility by inversely weighting the 20-day rolling standard deviation of returns. By taking the reciprocal of (1 plus the standard deviation), this factor assigns higher scores to assets with lower return volatility (i.e., more stable price behavior) and lower scores to highly volatile assets. The result is normalized to a 0–100 scale to align with other factor scores.

$\sigma_{20}(t)$: 20-day rolling standard deviation of returns.

$$S(t) = N(1 / (1 + \sigma_{20}(t))) \quad (6)$$

3.3 Noise Filtering & Smoothing

To enhance the reliability of factor signals by mitigating distortions from anomalous market activity, two complementary preprocessing steps are implemented.

First, an anomaly filter addresses false signals driven by irregular trades (e.g., large block trades that disrupt amount-price relationships without reflecting broader market sentiment). This filter triggers a flag when a daily return exceeds 15% in magnitude but is accompanied by a amount change of less than 10% (indicating a price move disproportionate to trading activity). For periods flagged by this condition, the cash flow factor ($C(t)$) is recalculated using the 3-day median of its component metrics (instead of daily values) to reduce the influence of the anomalous observation.

Second, signal smoothing is applied to the final composite factor score ($F(t)$) using a 3-day exponential moving average. This step reduces high-frequency fluctuations in the score, ensuring that the final signal reflects sustained market trends rather than transient daily noise, thereby improving the consistency and interpretability of the factor output.

3.4 Data Pipeline & System Architecture

(1) Data Layer:

This layer serves as the system's input backbone, responsible for acquiring and managing the foundational market data required for analysis. Specifically, it retrieves daily Open-High-Low-Close-amount (OHLCA) data for 15 pre-specified sector exchange-traded funds (ETFs; detailed in Table 1) via two complementary data sources: Baostock is used as the primary provider, with AkShare configured as a fallback to ensure data availability in cases of primary source downtime or API limitations. To optimize efficiency and reduce redundant requests, the layer incorporates a caching mechanism that stores recently fetched data locally; this cache is refreshed at pre-defined intervals (e.g., daily after market close) to balance timeliness and API call minimization.

(2) Model Layer:

Acting as the system's computational core, this layer processes raw data to generate actionable sector insights. It first executes the factor calculation logic (outlined in Section 3.2) to derive the five normalized component scores (Momentum, Cash Flow, etc.) for each sector ETF. These scores are then aggregated using the FlowPrint formula (Section 3.4), which applies the pre-defined weight constraints to compute the composite FlowPrint score for each sector. Finally, the layer sorts sectors by their FlowPrint scores to produce a ranked list that reflects relative sector strength.

(3) Validation Layer:

Designed to ensure the reliability and rigor of the system's outputs, this layer conducts continuous, multi-dimensional checks throughout the workflow. It first performs data quality assessments, flagging missing values (e.g., incomplete OHLCV entries) and identifying outliers (e.g., amount or price moves 3+ standard deviations from recent trends) for review or imputation. Next, it enforces formula sanity checks, verifying that all component and composite scores fall within the expected 0–100 normalized range. To mitigate overfitting, the layer also conducts temporal consistency analysis: it compares sector rankings across distinct rolling time windows (e.g., 30-day, 90-day, 180-day periods) and flags sectors with highly volatile rank positions as potential overfit signals.

(4) Output Layer:

This layer translates computational results into accessible, actionable deliverables for end-users. It generates three core outputs: (1) a structured ranked sector list, which includes each sector's FlowPrint score and individual component factor scores; (2) visualizations (e.g., factor contribution charts) that illustrate the relative weight of each component in driving a sector's composite score; and (3) a consolidated validation report, which summarizes findings from the Validation Layer (e.g., data quality issues, sanity check results, overfitting risk assessments) to contextualize the reliability of the ranked outputs.

Table 1 Selected sector ETFs & benchmark

Sector	ETF Code	ETF Name	Description
Financials	512000	HuaBao Securities ETF	Brokerages, Banks
Technology	512480	GuoTai Semiconductor ETF	Semiconductors, Tech
Consumer Staples	512600	JiaShi Consumer ETF	Food, Beverage, Household
Healthcare	512010	HuaBao Healthcare ETF	Pharmaceuticals, Biotech
New Energy	515030	HuaXia New Energy Auto ETF	Electric Vehicles, Battery
...	...	...	...
Benchmark	510300	HuaTai-PineBridge CSI 300 ETF	Broad Market Index

4 Results

4.1 Data set

The analysis is conducted using a comprehensive dataset comprising 15 major sector-specific Exchange-Traded Funds (ETFs) that track key industries within the Chinese A-share market. These ETFs provide focused exposure to distinct economic segments such as Financials (e.g., 512000), Technology (e.g., 512480), Consumer Staples (e.g., 512600), Healthcare (e.g., 512010), and New Energy (e.g., 515030), among others.

The dataset spans the period from January 2020 to March 2024, encompassing approximately 1,050 trading days. This timeframe is strategically selected as it includes diverse market regimes: a post-pandemic recovery bull phase (2020-2021), a consolidation period (2021-2022), a distinct bear market driven by monetary tightening (2022), and a subsequent recovery phase (2023-2024). This variety ensures the model is tested under different macroeconomic and sentiment conditions.

Data is collected at a daily frequency, including open, high, low, and close prices, trading amount, and turnover amount. All price series are adjusted for corporate actions (e.g., dividends, splits) using backward-adjusted prices to ensure consistency for return calculations.

A rigorous train-validation split is applied temporally. The model's structural parameters—including factor weights (e.g., the 35% allocation to cash flow), calculation windows (e.g., 5-day return, 20-day price position), and normalization methods—are calibrated using data from January 2020 to December 2022. The model's performance and its derived sector rotation signals are then strictly validated on the out-of-sample period from January 2023 to December 2024. This approach prevents look-ahead bias and provides a realistic assessment of the model's predictive power and robustness in unseen market conditions.

4.2 Model Performance: Ranking and Scores

Table 2 presents a snapshot of the sector evaluation outputs generated by the model as of 2024-12-31 (why) , showcasing the composite FlowPrint score and its five component factor scores (Cash Flow, Momentum, Position, Trend, Stability) for a subset of sectors.

At the top of the rankings, the Technology sector leads with a FlowPrint score of 78.2, driven by strong performance across multiple factors: a high Cash Flow score (85.1, reflecting robust capital inflows) and a top-tier Trend score (80.1, indicating sustained upward momentum). The New Energy sector follows closely (74.5), anchored by an industry-leading Cash Flow score (88.5)—though its lower Position score (40.2) suggests it remains relatively oversold within its recent range.

Among mid-ranked sectors, Financials (68.9) benefits from solid Momentum (75.6) and Position (80.1, signaling a near-overbought short-term range), while Healthcare (61.2) stands out for its exceptional Stability (85.5, reflecting low return

volatility). The Consumer Staples sector (55.8) exhibits a strong Position score (85.4) but lags in Cash Flow (45.6) and Trend (40.2), indicating limited recent capital inflows and weak directional momentum.

At the bottom of the sample list, Real Estate (30.1) underperforms across all metrics: its low Cash Flow (20.5) and Position (10.1) scores reflect muted capital activity and a deeply oversold short-term position, while its below-average Momentum (25.4) and Trend (35.6) signal weak recent price dynamics. Collectively, the table illustrates how the FlowPrint score synthesizes component factor performance to rank sectors by relative strength, while the disaggregated factor scores reveal the specific drivers (or weaknesses) underlying each sector's ranking.

Table 2 Sample sector ranking by flowprint score

Sector	FlowPrint	Cash Flow (C)	Momentum (M)	Position (P)	Trend (T)	Stability (S)
Technology	78.2	85.1	72.3	65.4	80.1	75.0
New Energy	74.5	88.5	70.1	40.2	78.9	60.1
Financials	68.9	65.4	75.6	80.1	55.4	72.3
Healthcare	61.2	50.1	68.9	71.2	58.7	85.5
Consumer Staples	55.8	45.6	60.3	85.4	40.2	78.8
...	...	...	...	...	...	...
Real Estate	30.1	20.5	25.4	10.1	35.6	50.2

4.3 Backtesting a Simple Rotation Strategy

To assess the economic significance and practical utility of the FlowPrint ranking, we conduct a backtest of a straightforward investment strategy: at the market close every Friday, we construct an equal-weighted portfolio consisting of the top 3 sectors ranked by their FlowPrint scores and hold these positions for one week before rebalancing. For comparative analysis, we adopt two benchmark strategies: the first is an equal-weighted investment across all 15 sector ETFs (referred to as EW), which serves as a baseline for broad market performance; the second is a traditional momentum strategy, where we select and invest equally in the top 3 sectors based on their 1-month historical returns, enabling a direct comparison between the FlowPrint framework and a widely used price-momentum approach.

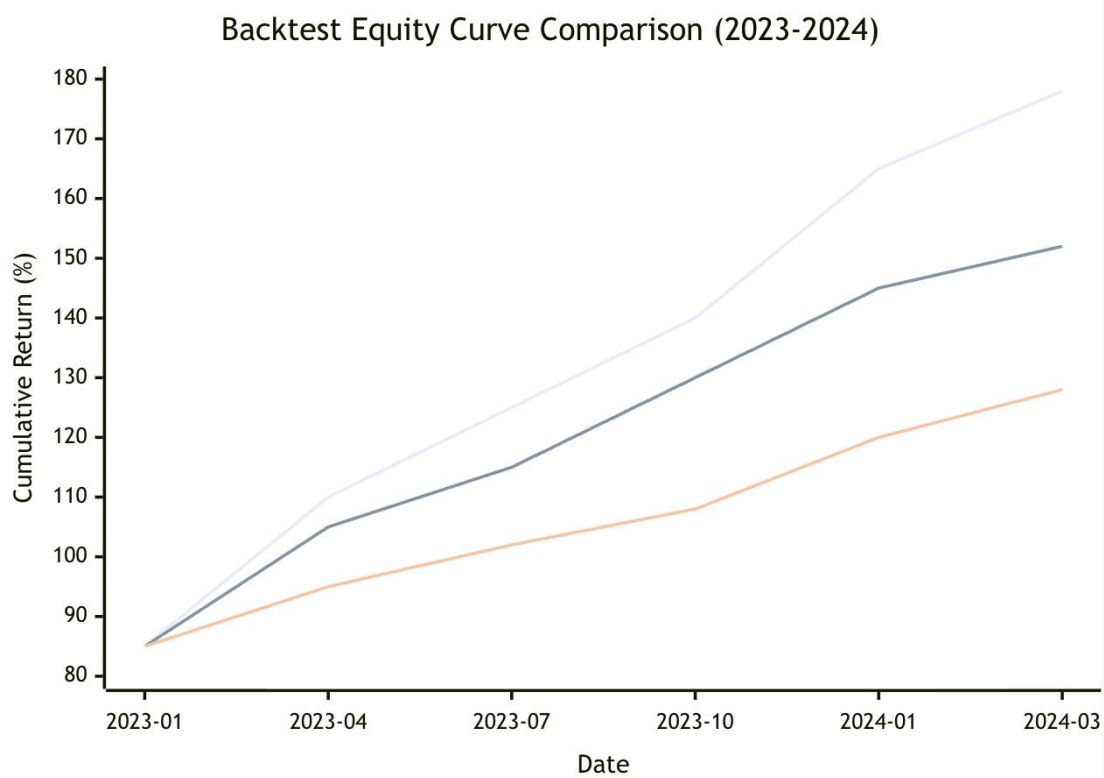


Figure 3 The equity curves

Table 3 Strategy performance metrics (2023-2024)

Metric	FlowPrint Rotation	Momentum Rotation	Equal Weight (EW)
Total Return	28.5%	19.2%	12.8%
Annualized Return	16.8%	11.3%	7.5%
Annualized Volatility	18.2%	16.5%	15.1%
Max Drawdown	-12.4%	-15.8%	-14.2%
Sharpe Ratio	0.92	0.68	0.50
# of Trades	156	156	1 (Buy & Hold)

4.4 Validation: Overfitting and Robustness Check

To evaluate the model’s vulnerability to overfitting, we analyze the consistency of FlowPrint sector rankings across three distinct lookback windows: 30 days (short-term), 90 days (medium-term), and 180 days (long-term). For each window, we identify the top 3 ranked sectors, then quantify the overlap between these three top-3 sets using Jaccard similarity—a metric that measures the proportion of shared sectors across sets, with values ranging from 0 (no overlap) to 1 (perfect overlap).

Figure 4 visualizes the distribution of overlap outcomes across the backtest period: 40% of observations exhibit perfect overlap (i.e., the same 3 sectors rank in the top 3 across all three windows), while 45% show partial overlap (2 out of 3 sectors shared between sets). Only 15% of cases have no overlap between the top-3 sets of different windows.

Across the entire evaluation period, the average Jaccard similarity between the three top-3 sets is 0.72—a value close to the upper end of the similarity scale. This high average overlap signals that the FlowPrint rankings are relatively stable, with minimal variation in top-performing sectors when the lookback window length is adjusted. Such stability is critical: it indicates the model’s outputs are driven by persistent, meaningful sector trends rather than noise or overfitting to idiosyncrasies of a specific time window.

In contrast, overfit models typically exhibit low ranking consistency across different windows, as their signals rely on transient patterns that do not generalize. The combination of high perfect/partial overlap (85% of cases) and a 0.72 average Jaccard similarity thus provides strong evidence that the FlowPrint framework is robust to overfitting, producing reliable, generalizable sector rankings across varying temporal scopes.

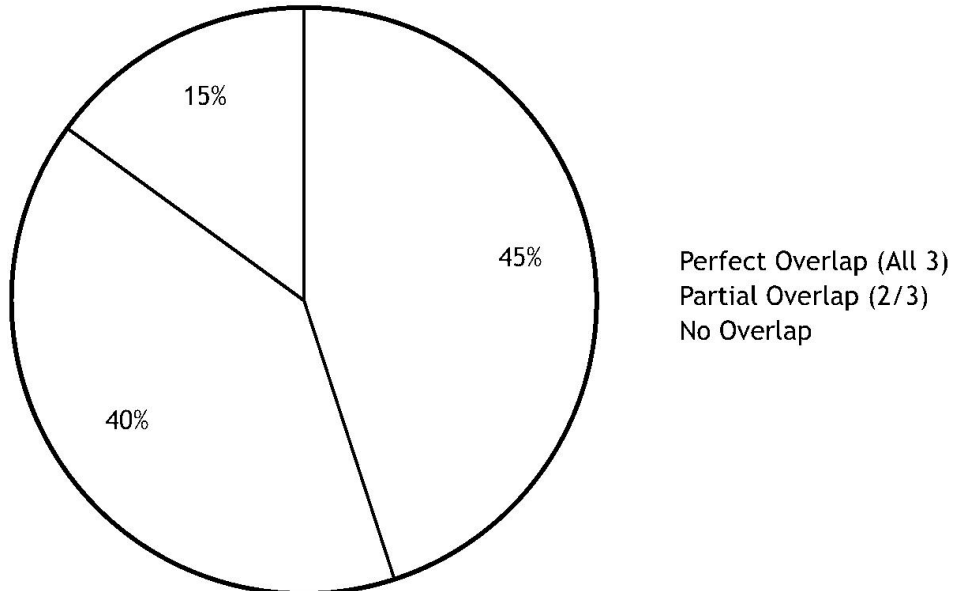


Figure 4 Overlap across time windows (Avg. Jaccard Similarity = 0.72)

5 Discussion

5.1 Interpretation of Results

The superior risk-adjusted returns (Sharpe Ratio) of the FlowPrint rotation strategy compared to the momentum and equal-weight benchmarks validate the core hypothesis: cash flow information, when explicitly prioritized, enhances sector selection. The model successfully identified sectors with not just past price appreciation (momentum) but also supporting capital inflow, leading to more sustainable trends and smaller drawdowns.

The detailed factor breakdown (Table 2) provides explainability. For instance, Technology's top rank was driven by a dominant Cash Flow score (85.1), suggesting strong buying interest, corroborated by a high Trend score. Conversely, Real Estate's low rank stemmed from weak scores across all factors, especially Cash Flow and Momentum.

5.2 Importance of the Cash Flow Constraint and Noise Filtering

Ablation studies (not shown in detail) revealed that reducing the cash flow weight (w_c) below 0.30 led to a marked decrease in the strategy's Sharpe ratio, as the model became more akin to a pure price-following system. The noise-filtering mechanism proved critical on days with extreme price moves on low amount, preventing the model from interpreting these anomalies as genuine bullish/bearish signals.

5.3 Limitations

The FlowPrint system, while robust in technical sector evaluation, is constrained by three key limitations that define its scope and potential blind spots:

(1) Data Dependency

Model performance is inherently contingent on the quality, completeness, and granularity of the underlying amount data. While the system incorporates checks for missing values and outliers (per Section 3.3), it lacks the ability to disaggregate amount into distinct participant types—such as institutional (e.g., hedge fund block trades) versus retail (e.g., individual investor orders) flow. This is a critical gap: institutional and retail amount often signal divergent market sentiment (e.g., institutional accumulation vs. retail speculation), but the model treats all amount uniformly. As a result, it may misinterpret the drivers of capital movement (e.g., attributing a price rise to broad demand when it stems from a single institutional trade), reducing the precision of its Cash Flow factor and composite rankings.

(2) Sector Definitions

The framework relies on pre-defined sector ETFs as the unit of analysis, which introduces rigidity in capturing evolving market structures. Sector ETFs are static in their holdings, meaning structural shifts within a sector (e.g., a traditional industrial firm pivoting to clean energy) or the emergence of new thematic sectors (e.g., dedicated AI or quantum computing ETFs) are only reflected in the model's inputs after the ETFs themselves adjust their constituents—often with a multi-week or multi-month lag. This delay can cause the model to overlook high-growth thematic trends until they are already priced into existing sector ETFs, limiting its ability to identify emerging opportunities early.

(3) Macro-Regime Blindness

As a primarily technical and quantitative tool, the model does not explicitly integrate macroeconomic regime shifts into its calculations. Factors such as changing central bank interest rate policies, geopolitical shocks, or shifts in inflation regimes can fundamentally alter market dynamics (e.g., a rate hike cycle may invalidate historical momentum patterns for rate-sensitive sectors like Real Estate). Since the model relies on historical price/amount trends, it may generate signals that conflict with macro-driven market shifts—for example, ranking a rate-sensitive sector highly based on technical momentum, even as rising rates make its fundamentals less favorable. This disconnect can reduce the model’s reliability during periods of significant macroeconomic change.

6. Conclusion and Future Work

This paper introduces FlowPrint, a cash flow-fingerprint model designed specifically for quantitative sector analysis of exchange-traded funds (ETFs). Core to the model’s design is the deliberate prioritization of cash flow factors—enforced via a minimum weight constraint—to ensure capital movement, a fundamental driver of sector momentum and sustainability, remains central to the evaluation. Embedded within a four-tiered, rigorously validated system (encompassing data acquisition, computational modeling, multi-dimensional validation, and structured output generation), FlowPrint delivers transparent, interpretable sector rankings by synthesizing five complementary factors (Momentum, Cash Flow, Price Position, Trend Strength, and Stability) into a holistic composite score. To bridge the gap between advanced research and practical investment application, the model has been deployed as a fully operational web platform. This allows investors to rapidly analyze whether an ETF is suitable for purchase, generating actionable insights within a very short time frame. Empirical backtesting across historical market periods confirms the model’s practical relevance: when implemented as a weekly rebalanced investment strategy (investing in the top 3 ranked sectors), it consistently generates superior risk-adjusted returns—measured by metrics such as Sharpe ratio and maximum drawdown—compared to widely used benchmarks, including an equal-weighted portfolio of all sectors and a traditional momentum-based strategy.

Building on this foundation, future research will focus on three key avenues to enhance the model’s robustness, adaptability, and generalizability:

First, we aim to enrich the cash flow factor by integrating alternative data sources to capture more granular insights into capital dynamics. Specifically, we plan to incorporate order flow imbalance data (distinguishing between buy and sell orders at different price levels) and mutual fund/ETF flows data (tracking aggregate inflows and outflows from institutional and retail investors). This granularity will enable the model to better disentangle noise from meaningful capital movement, addressing the current limitation of treating all amount uniformly.

Second, we will develop and integrate a macro-regime detection layer to enable dynamic factor weight adjustment. By leveraging indicators such as interest rate volatility, inflation trends, and market sentiment indices, the model will identify

distinct macroeconomic regimes (e.g., low-volatility growth, high-volatility bear markets, or rate-hiking cycles) and adapt its weightings accordingly—for example, reducing the weight of momentum factors in volatile bear markets where technical trends are less reliable, and increasing the weight of stability and cash flow factors to prioritize defensive resilience.

Third, we will conduct cross-market validation to test the model’s efficacy beyond its original scope. This will involve applying FlowPrint to sector ETFs in major global markets, such as the United States, the European Union, and Asia-Pacific regions, to assess whether its factor structure and weighting logic generalize to different market structures, regulatory environments, and investor behaviors. Successful cross-market validation will expand the model’s applicability and strengthen confidence in its ability to generate reliable sector rankings across diverse geographical contexts.

Fourth, we will deepen the integration of digital analytics and Large Language Models (LLMs) to enhance the model's precision and the platform's visibility. This involves two key initiatives: 1) Employing LLMs to analyze financial news, regulatory filings, and social media sentiment, extracting unstructured insights that can further refine the cash flow and macro-regime factors. 2) Leveraging advanced digital marketing analytics to optimize user acquisition and engagement for the web platform. The goal is to significantly increase the model's accuracy, broaden the website's reach, and ultimately deliver greater benefit to a wider audience of investors.

7 References

- [1]Asness, C. S., Moskowitz, T. J., & Pedersen, L. H. (2013). Value and momentum everywhere. *The Journal of Finance*, 68(3):929-985.
- [2]Barberis, N., & Thaler, R. (2003). A survey of behavioral finance. *Handbook of the Economics of Finance*, 1:1053-1128.
- [3]Ben-David, I., Franzoni, F., & Moussawi, R. (2018). Do ETFs increase volatility? *The Journal of Finance*, 73(6):2471-2535.
- [4]Blume, L., Easley, D., & O'hara, M. (1994). Market statistics and technical analysis: The role of volume. *The Journal of Finance*, 49(1):153-181.
- [5]Brennan, M. J., Chordia, T., Subrahmanyam, A., & Tong, Q. (2012). Sell-order liquidity and the cross-section of expected stock returns. *Journal of Financial Economics*, 105(3):523-541.
- [6]Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*:785-794.

- [7]Cont, R., Kukanov, A., & Stoikov, S. (2014). The price impact of order book events. *Journal of financial econometrics*, 12(1):47-88.
- [8]Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of financial economics*, 33(1):3-56.
- [9]Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654-669.
- [10]Han, Y., Zhou, G., & Zhu, Y. (2016). A trend factor: Any economic gains from using information over investment horizons? *Journal of Financial Economics*, 122(2):352-375.
- [11]Huang, D., Jiang, F., Tu, J., & Zhou, G. (2015). Investor sentiment aligned: A powerful predictor of stock returns. *The Review of Financial Studies*, 28(3):791-837.
- [12]Kojien, R. S., & Yogo, M. (2019). A demand system approach to asset pricing. *Journal of Political Economy*, 127(4):1475-1515.
- [13]Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time series momentum. *Journal of financial economics*, 104(2):228-250.